Extracting Roads from Medium Resolution Satellite Imagery

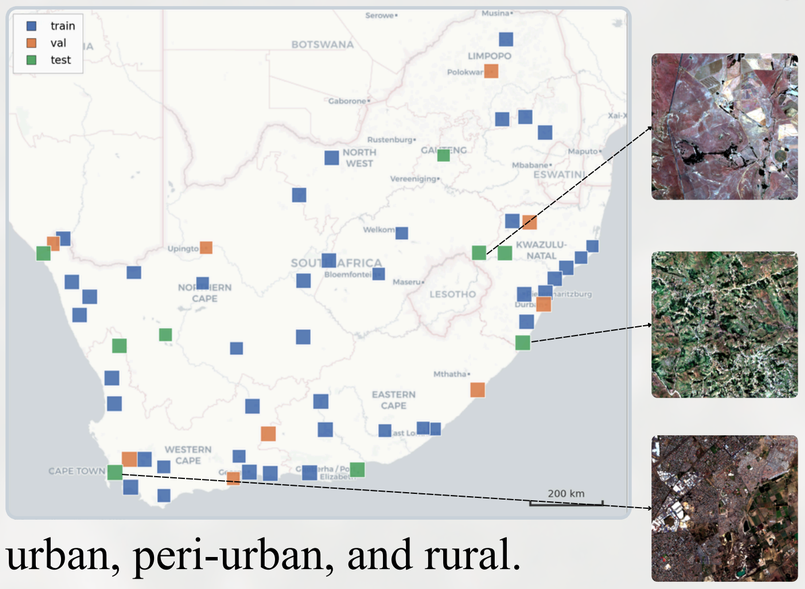
1. Introduction

We develop AI tools to monitor how road networks change over time, based on satellite imagery. This supports urban planning and assessing the environmental impact of road networks. While Satellite imagery makes this possible at scale, high-resolution imagery is expensive, and licensing often restricts model training. To overcome this, we use freely available Sentinel imagery, which covers the globe every few days. This data is low resolution however, and this needs to be accounted for in the algorithms.

2. The Dataset

Cloud-free images with road labels were carefully curated to build a public dataset covering diverse locations in SA.

The locations were stratified across 9 biomes and the 3 levels of urbanisation: urban, peri-urban, and rural.



3. Road Extraction Approaches using Sentinel Imagery

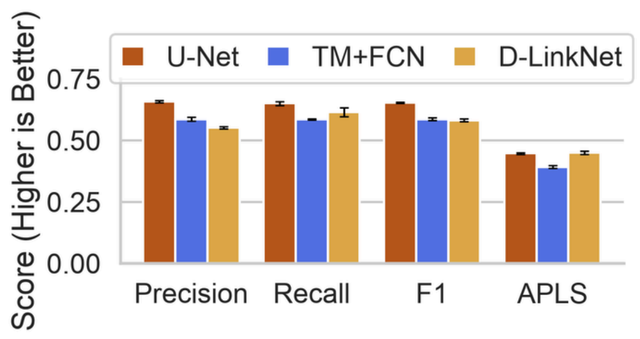
Is a Larger Model Better?

Large TerraMind (TM) model is:

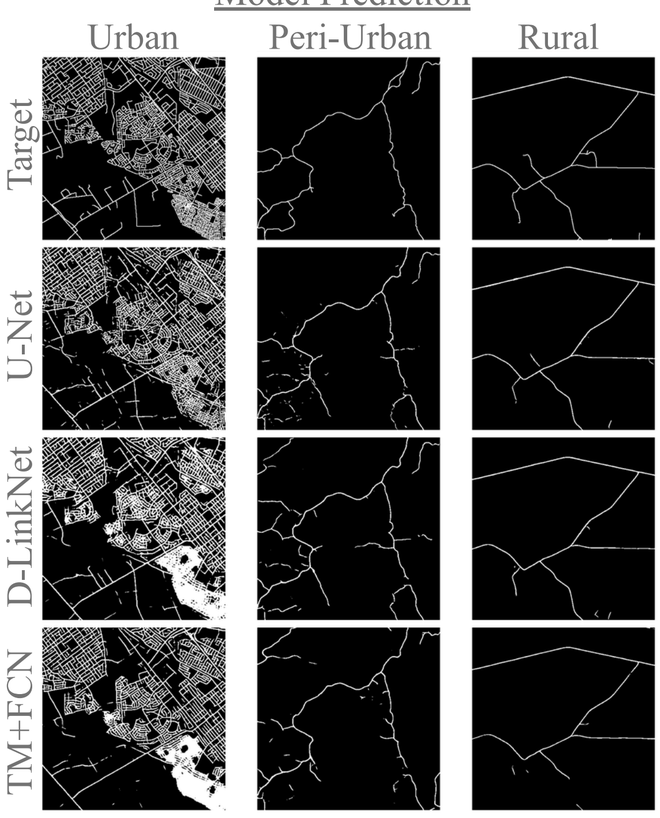
(1) **adapted:** Best adapted TM model is fully fine-tuned with FCN Decoder. Trying to fine-tune "smarter" with LoRA (fine-tunes less parameters) was ineffective.

(2) **compared** to medium-sized models U-Net and D-LinkNet

Results



Model Prediction



TM cannot match U-Net & only sometimes matches D-LinkNet for road segmentation. Possible reasons:

(1) TM has to learn certain assumptions useful for the task while the medium-sized models have them built into their design

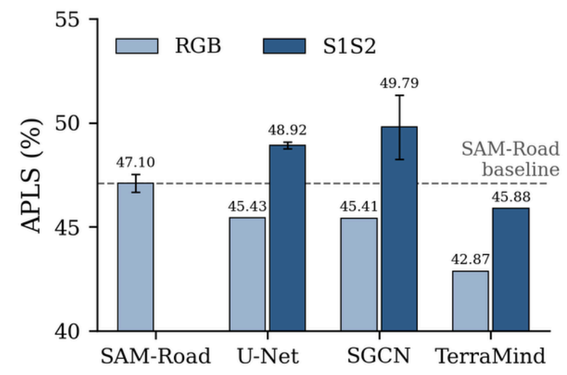
(2) TM sees 8x more coarse info than U-Net.

Graph-Based Road Extraction

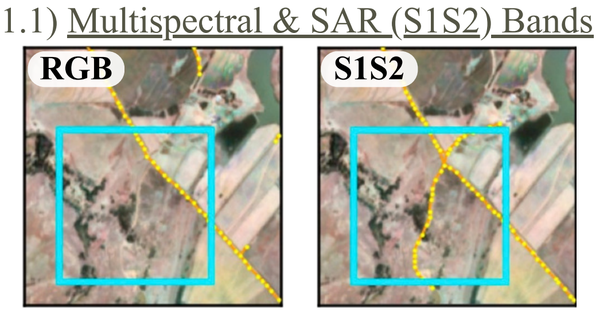
Road networks are better represented as graphs. We explore modifications to SAM-Road, a model that predicts road graphs directly.

SAM-Road Model Modifications

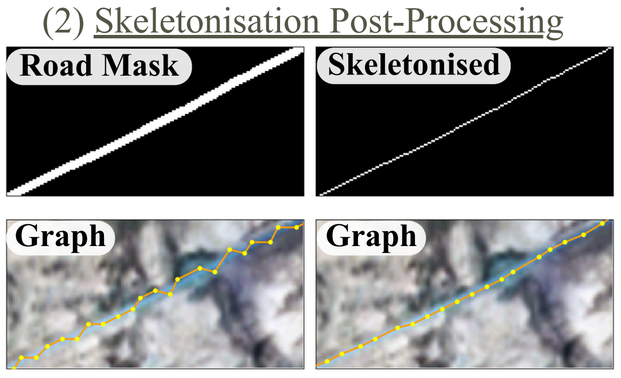
(1) Modify Segmentation Component



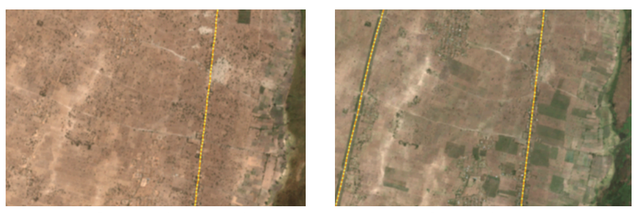
(1.1) Multispectral & SAR (S1S2) Bands



(2) Skeletonisation Post-Processing



Integration into web application for practical applications



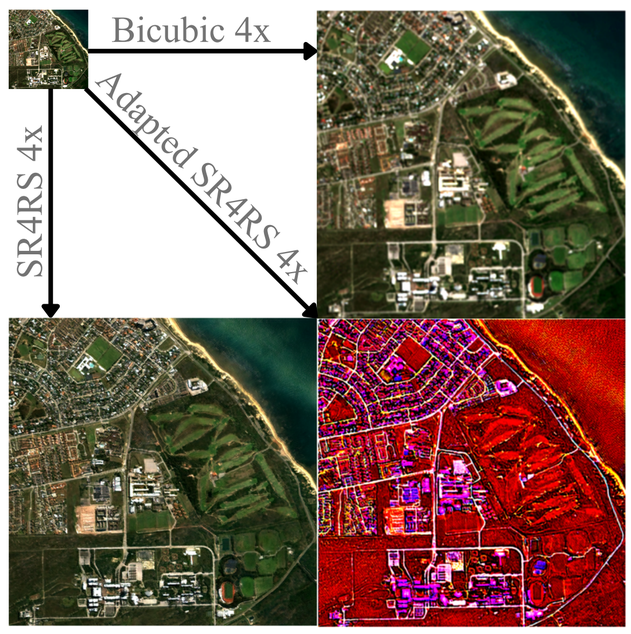
Before (2015) After (2025)

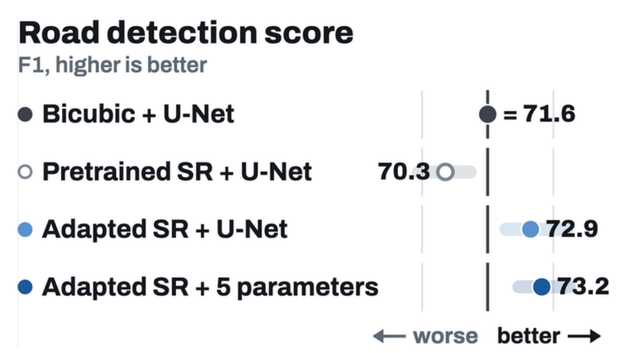
Additional bands are beneficial, which our modifications can utilise with negligible additional cost. Our work offers a practical foundation for monitoring road networks.

Make the data better - with Super Resolution

Idea: 10 m is too coarse, so sharpen the image first with super-resolution (SR).
Problem: Off-the-shelf SR makes road maps worse.
Fix: Train the SR together with the road detector. This wins back the lost accuracy.
Extension: Swap the 24M-parameter U-Net for just 5 parameters. It still finds roads more accurately than bicubic + U-Net.

Image Upsampling Result





We show that adapting the SR with either U-Net or a 5 parameter logistic head beats the standard pipeline at marking road pixels (higher F1). The logistic head is much cheaper to train, and attained F1 comparable to the U-Net variant. However, the logistic head degrades the connectivity of roads predicted.

Team Members
Kelvin Wei - WXXKEL001
Jing Yeh - YHXJIN001
Catherine Li - LXXCAT004

Supervisors
A/Prof. Patrick Marais
A/Prof. Jonathan Shock

University of Cape Town
Department of Computer Science
Cape Town, South Africa